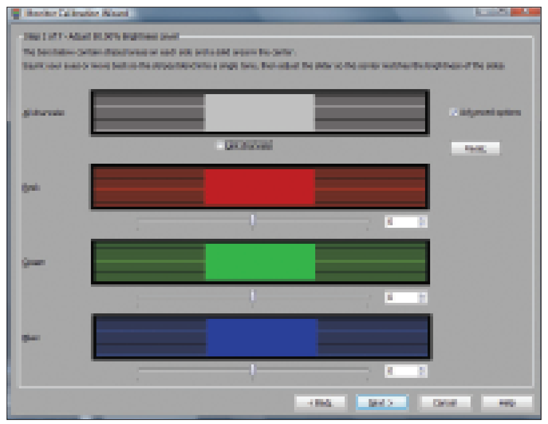
using a suitable calibration device such as a colorimeter and adjusting the physical controls on the display until the measurements reach the required levels. This sets the display to a known and repeatable state so that successive profiles are always created from the same starting point.

The input to output transfer characteristics of displays vary, two examples of the same model LCD or CRT monitor may not produce exactly the same brightness or colour when driven by identical video signals. Most computer monitors do not have individual colour channel linearity controls. Uncorrected monitors not only deviate from the ideal desired transfer curve, but are also likely to have different transfer characteristics for each colour channel. If the Red, Green and Blue channels do not track together the display will exhibit unwanted, shade dependant hue variations. Poor RGB tracking is most apparent where obvious colour tints appear in monochrome images with a wide range of greys.

Software Display Calibrators

Although the approach has been abandoned in Adobe Photoshop CS4 and Adobe now specifically recommend using a third party hardware based display calibrator, earlier versions of Photoshop included a software only display calibrator and profiler applet called Adobe Gamma. The Photoshop install places this in the Windows Control Panel and also installs a short cut in the Start folder to the Adobe profile loader.



The Corel Paint Shop Pro 50% brightness profiling display. The slider values are adjusted until the solid centre colour appears to be the same brightness as the line patterns on either side

Adobe Gamma attempts to calibrate and profile a display by eye. First a graduated brightness scale is used to calibrate the brightness and contrast settings then a series of solid colour / alternate line patterns is displayed. The center solid patch is adjusted in brightness by eye until it matches the brightness of the alternate lines. Required corrections are then derived from these relative settings and used to construct a profile. Corel Paint Shop Pro Photo X2 has a similar utility, but improves accuracy by providing adjustment at 5 different brightness levels. Adobe Gamma only corrects at a single brightness level. These 'eyeball' calibrators are perhaps better than nothing, but are not as accurate as the current hardware calibrators.

Recent display drivers include controls which can be used to modify display characteristic, such as gamma controls. These should be turned off, or left at default values, before (and after) a display is calibrated and profiled.

Boot Time Display Profile Loaders

The display colour response is not corrected by making adjustments to the display itself but by applying corrections to the graphics card. Usually these corrections are loaded by a profile loader applet loading the correction values from the display profile into the colour Look Up Table (LUT) on the graphics card at boot time. These corrections are therefore global, affecting everything displayed, following boot up.

Most colour profiling applications are supplied with a profile loader applet which is installed during installation of the main program. A short cut to the profile loader is placed in the Start folder and is therefore called during operating system boot. Under Windows Vista these can be found via `Start > All Programs` and opening the Startup folder. Any profile loaders should be listed here.

If several applications with colour management that support profile boot loaders are installed on the same PC there can be some confusion. The LUT will eventually be loaded by whichever loader and profile is loaded last in the boot sequence and this may not be the desired profile. Users should manually edit the Start folder and remove all loader short cuts except to the desired loader. Ideally this should be for a hardware display calibrator and its associated software.

Hard Copy Colour Proofing

In theory the ideal colour proof work flow is to print proofs on the printer that will be used for production runs. In the case of commercial press printing this is impractical because the presses involved are not designed to print just the few pages required for colour proofing. In the past hard copy press print proofs were produced using a (DuPont) Chromalin, (Kodak) Matchprint or similar proofing system. Unfortunately the word 'Chromalin' is often used like the word 'Hoover' is used as a generic term for vacuum cleaners and if you ask a printer for a Chromalin proof that may not literally be what you get.

By the time pages reach the printing press the edited page layouts have been split into the component C, M, Y and K (known as 'separations') used to print the finished pages. To make a Chromalin proof, the separations, either in digital file form, or as page spread size photographic negatives are used to produce positive colour exposures of the four colours on four sheets of plastic. These sheets are then aligned and bonded together to form the final colour proof. Other photographic proofing systems use the separation negatives to produce a colour photographic print on photo print paper. The digital files or negatives used to make the proof are also used to make the printing plates for the press, so in principle there is an exact correspondence between the proof and the press print.

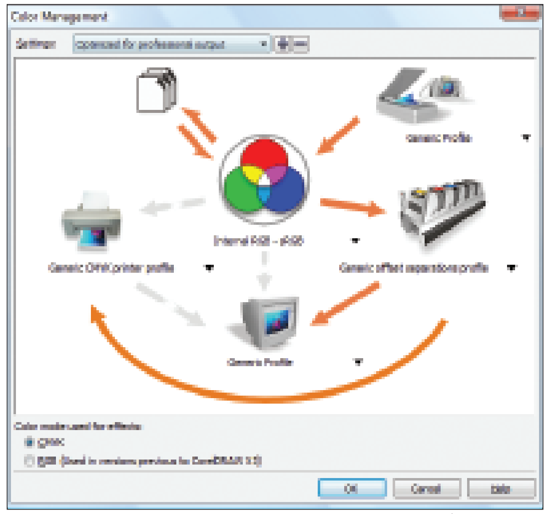
The problem with all these colour press proofing systems is that they do not use the inks and paper that will be used in the final print run. They will show errors such as registration problems or obvious colour errors introduced at the separation stage, but will not necessarily produce exactly the same colour appearance as the final printed output. This is particularly obvious with proofs printed onto photographic paper, as the white of the photo print paper normally won't be the same as the white of the paper used in the press.

Hard copy colour proofs are relatively expensive and time consuming to produce, commercial press operators or print service bureaus apply a charge for each colour proof (in the UK, £45 to 50 each for A4 proofs). So publications will normally only request hard copy colour proofs for images of particular importance.

Soft Proofing

Soft proofing attempts to use either a computer display or a desktop colour inkjet or laser printer to emulate the performance of the final output media, i.e. a printing press.

It is cheap, convenient and fast. However the term 'soft proofing' is also used to refer to colour proofs, such as PDF files sent by email, that do not attempt accurate colour reproduction, but serve merely as general proofs for checking layout, text and overall appearance.



The CorelDRAW Color Management control panel with the base settings for soft proofing selected. Both the display and the desktop printer will show the image as it would appear on a commercial printing press. In this example all the individual profiles have been left at the starting generic setting